

Jiayu Xu

PhD Candidate in Chemical Engineering

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Research Interests

Discrete element method (DEM) and GPU parallel computing
 Non-spherical particles and multiphase-flow simulation
 CFD-DEM coupling and multiscale particle-fluid modelling
 AI-assisted particle identification, tracking, and parameter inference
 Numerical modelling of industrial particulate processes

Education

PhD Candidate	University of Chinese Academy of Sciences , Chemical Engineering • Institute of Process Engineering, Chinese Academy of Sciences • GPA: 3.70/4.00 • Advisors: Prof. Ji Xu and Prof. Wei Ge • Research on DEM, GPU computing, non-spherical particles, multiphase-flow simulation, and AI-assisted particle analysis	Beijing, China Sept 2021 – present
Bachelor of Engineering	Xiangtan University , Chemical Engineering and Technology • School of Chemical Engineering • GPA: 3.73/4.00 • Advisor: Prof. Yefeng Zhou • Research on CFD-DEM coupling, gas-solid fluidized beds, wet granular flows, and cohesive particle modelling	Xiangtan, China Sept 2017 – June 2021

Research Experience

Institute of Process Engineering, Chinese Academy of Sciences , Gas-Solid Multiphase Flow and CFD-DEM Coupling • Developed CFD-DEM coupled solvers for gas-solid fluidized beds, fluidized-bed reactors, and particle conveying systems • Investigated multiscale particle-fluid methods including CG-DPM and MP-PIC	Beijing, China Jan 2025 – present
Institute of Process Engineering, Chinese Academy of Sciences , Large-Scale DEM Simulation and High-Performance Computing • Developed CUDA/MPI-based multi-GPU DEM algorithms for complex polyhedral particle systems • Modelled concave and convex polyhedra and superquadric particles • Applied the developed methods to rotary drums, hoppers, fixed beds, and granular piles • Developed high-performance numerical software for large-scale particulate systems	Beijing, China Sept 2022 – Mar 2026
Institute of Process Engineering, Chinese Academy of Sciences , AI-Assisted Particle Modelling and Parameter Identification • Developed deep-learning methods for irregular-particle image segmentation • Reconstructed particle velocity fields from experimental images • Predicted DEM model parameters through simulation-experiment integration	Beijing, China July 2024 – Feb 2026

Xiangtan University, Porous Carbon Preparation and CO₂ Adsorption

- Investigated porous carbon preparation from chitosan-derived hydrochar
- Analysed the effects of KOH activation and hydrothermal additives on pore structure and CO₂ capture performance

Xiangtan, China

Sept 2019 – June 2021

Selected Projects

Quantitative Investigation of Discharge Rate in a High-Temperature Rotary Feeder

May 2024 – May 2026

Industrial research project with Baosteel

- Developed a particle-flow model for a rotary feeder
- Analysed the effects of feeding mode, filling level, rotational speed, and outlet geometry
- Supported structural optimisation and operating-parameter selection

DEM Modelling of Wet-Particle Discharge from a Hopper

June 2022 – Aug 2024

- Developed a wet-particle DEM model for hopper discharge
- Investigated the effects of formulation, moisture content, and process geometry
- Supported formulation optimisation and process scale-up

Quantitative Study of a High-Temperature Charging System

Sept 2019 – Nov 2021

Industrial research project with Baosteel

- Constructed charging and burden-distribution models
- Analysed burden trajectory, surface profile, ore-to-coke ratio, and particle-size distribution
- Supported operating optimisation and reduction of coke consumption

Selected Publications

DEM framework for large-scale concave and convex polyhedral particle systems with multiple GPUs

Jan 2026

Jiayu Xu, Ji Xu, Wei Ge, et al.
(Computational Mechanics)

Large-scale multiple-GPU DEM simulation of polyhedral particle systems

Jan 2026

Jiayu Xu, Ji Xu, Wei Ge, et al.
(Powder Technology)

An efficient non-spherical particle tracking strategy based on deep-learning and simulation-experiment integration

Jan 2026

Jiayu Xu, Shuai Zhang, Wei Ge, et al.
(Powder Technology)

Deep-learning-based property parameters identification for DEM simulations

Jan 2026

Jiayu Xu, Ji Xu, Wei Ge, et al.
(Chemical Engineering Science)

Patents and Software

- Invention patent for target-object detection methods, devices, electronic equipment, and storage media
- Invention patent for multi-GPU parallel computation of complex polyhedral particle systems
- Invention patent for multiphase-flow solid-fluid coupling methods
- polyDEM V1.0 — complex-particle DEM simulation software, Registration No. 2025SR2431344
- DEMms V8.0 — GPU-based DEM software, Registration No. 2026SR0445479

Skills

Programming: C/C++, Python, MATLAB, CUDA, Shell

Parallel Computing: GPU computing, MPI, CPU/GPU heterogeneous computing, and domain decomposition

Numerical Methods: DEM, CFD-DEM, CG-DPM, MP-PIC, and MPS

Artificial Intelligence: Machine learning, deep learning, image segmentation, particle tracking, and CMA-ES optimisation

Software: Linux, Vim, Origin, AutoCAD, Aspen Plus, and Ansys Fluent

Honors and Awards

- Outstanding Graduate of Xiangtan University, 2021
- Zhanchen Enterprise Scholarship, 2020
- First-Class Scholarship of Xiangtan University, 2018, 2019, and 2020
- Third Prize, National College Chemical Engineering Design Competition, 2018